AAKARSH ANAND

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Google Scholar · GitHub · LinkedIn

Machine learning researcher working on foundation models and statistical genetics for health data

EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, CA
Ph.D., Computer Science · Advisor: Sriram Sankararaman — 2024 – 2029 (expected)
Major field: Artificial Intelligence · Minor fields: Data Science Computing, Computational Systems Biology
M.S., Computer Science (earned during the Ph.D.) — 2026
B.S., Computer Science — 2020 – 2024

EXPERIENCE

Graduate Student Researcher, Sankararaman Lab, UCLA (undergraduate researcher 2022–24) — Jan 2022 – present
Foundation models for health data, with Yuzhe Yang's lab, and statistical genetics methods at biobank scale.

Software Engineering Intern, LabCorp, Burlington, NC — Jun – Sep 2023
Reproducible pipelines for high-volume genetic sequencing data.

Research Intern (NSF REU), Bruins in Genomics Summer Program, UCLA — Jun – Sep 2022
Symbolic regression to explain gene–gene interactions learned by nonlinear models.

Research Assistant, Boutros Lab, UCLA — Jan 2021 – Apr 2022
Added the HISAT2 aligner to the lab's DNA alignment pipeline, which is now part of Metapipeline-DNA.

PUBLICATIONS

* equal contribution

Peer-reviewed

- Inertia-1: An Open Exploration of Wearable Motion Foundation Models
  Z. Xu*, **A. Anand***, S. Jiang*, C. Zhuang, Z. Shuai, S. Sankararaman, Y. Yang · **NeurIPS 2026 main track** · also an oral at the GenAI for Health workshop · code
- A biobank-scale method for learning modulators of gene-environment interaction underlying human complex traits from multiple environmental exposures
  Z. Liu, A. Ramteke, **A. Anand**, A. Gorla, M. Jeong, S. Sankararaman · **Genome Research** 2026 · selected from RECOMB 2026 · code
- Metapipeline-DNA: A comprehensive germline and somatic genomics Nextflow pipeline
  Y. Patel, C. Zhu, T. N. Yamaguchi, N. K. Wang, ..., **A. Anand**, ..., P. C. Boutros (43 authors) · **Cell Reports Methods** 6(3):101340, 2026
- A biobank-scale test of marginal epistasis reveals genome-wide signals of polygenic interaction effects
  B. Fu, A. Pazokitoroudi, Z. Shi, A. Kar, A. Xue, **A. Anand**, P. Anand, Z. Liu, R. Border, P. Pajukanta, N. Zaitlen, S. Sankararaman · **Nature Genetics** 57(12):3175–3184, 2025 · code · covered in a UCLA Health news release
- A scalable adaptive quadratic kernel method for interpretable epistasis analysis in complex traits
  B. Fu*, P. Anand*, **A. Anand***, J. Mefford, S. Sankararaman · **Genome Research** 34(9):1294–1303, 2024 · selected from RECOMB 2024 · code

In revision

- Disentangling phenotype scale, amplification, and effect heterogeneity underlying polygenic GxE architecture of complex traits
  Z. Liu, **A. Anand**, X. Qie, M. Costantino, M. Jeong, A. Gorla, N. Zaitlen, A. Dahl, S. Sankararaman · in revision at **Nature Genetics** · code
- Comprehensive gene heritability estimation reveals the genetic architecture of rare coding variants underlying complex traits
  Z. Liu, B. Fu, M. Jeong, P. Anand, **A. Anand**, S. Jang, A. Gorla, J. Zhu, P. Pajukanta, P. F. Palamara, N. Zaitlen, R. Border, S. Sankararaman · bioRxiv 2025 · in revision at **Nature Genetics**

CURRENT RESEARCH

- Disease prediction and genetics from wearable foundation models in five cohorts, with early work pairing them with language models (follow-up to Inertia-1, co-first author)
- A training loss that encourages a model's learned traits to be heritable and genetically correlated with disease
- A genome-wide test for gene–gene interactions in rare disease, supported by an Anthropic AI for Science grant
- A model of how genetic effects on gene expression vary across cell types and cell states in single-cell data
- An atlas of genetic correlation across 3,570 UK Biobank trait pairs in 53 tissues, with two collaborators

TALKS AND POSTERS

- Oral: Inertia-1, at The Third Workshop on GenAI for Health, NeurIPS 2026, Sydney — Dec 2026
- Explaining potential epistasis in genomic data using symbolic representations of complex black box models
  A. Anand, P. Anand · Bruins in Genomics Summer Program, UCLA, August 2022

AWARDS, FELLOWSHIPS, AND GRANTS

Anthropic AI for Science Grant, Rare Disease Research — 2026
NIH T32 Predoctoral Fellowship, UCLA Genomic Analysis Training Program (NHGRI T32 HG002536) — 2026 – 2027
Warren Alpert Computational Biology and AI Network Fellowship, UCLA — 2024

TEACHING

Teaching Assistant, UCLA — 2025 – 2026

- Machine Learning Applications in Genetics (CS C124/224) — Spring 2026
- AI for Computational Genomics (CS CM121/221, ~120 students) — Spring 2025
- Machine Learning Applications in Biomedicine (DS BMED 205) — Winter 2025